CSE400 – Fundamentals of Probability in Computing

Lecture 21 & 22: Inequalities and Tail Bounds (Fully Expanded Exam Scribe)

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1. Tail of a Random Variable

Definition

The tail of a random variable X is defined as:

$$P(X > x)$$

This represents the probability that X takes values larger than x.

Why do we care about tails?

- Jobs delayed more than 24 hours
- Packets dropped due to full buffer

These correspond to rare but critical events in systems.

Key Idea

- Mean → average behavior (easy to compute)
- Tail → rare extreme events (hard to compute)

Why are tails hard to compute?

- Requires summing many small probabilities

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2. Tail Example 1 (Binomial Case)

Problem

Suppose you're distributing n jobs among n servers at random. What is the probability that a particular server gets k jobs?

Intuition

- Jobs are assigned randomly
- Most servers get few jobs
- Occasionally, one server gets many jobs

This corresponds to a tail event (rare overload).

Model

$$X \sim \text{Binomial}(n, p)$$

Exact Tail Probability

$$P(X \geq k) = \sum_{i=k}^{n} \binom{n}{i} p^i (1-p)^{n-i}$$

Step-by-Step Logic

1. We are interested in the probability that X is at least k.
2. This includes all outcomes where X = k, k+1, ..., n.
3. The probability of exactly i successes is:
$$\binom{n}{i} p^i (1-p)^{n-i}$$
4. Therefore, summing over all i k gives the tail probability.

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3. Tail Example 2 (Poisson Case)

Problem

Jobs arrive to a datacenter according to a Poisson process with rate jobs/hour. What is the probability that k jobs arrive during the first hour?

Intuition

- Jobs arrive randomly over time
- Most hours have moderate arrivals
- Sometimes, a burst of arrivals occurs

This is a tail event (sudden spike in load).

Model

$$X \sim \text{Poisson}(\lambda)$$

Exact Tail Probability

$$P(X \geq k) = \sum_{i=k}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!}$$

Step-by-Step Logic

1. Poisson gives probability of exactly i arrivals.
2. Tail event requires summing probabilities from k to infinity.

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4. Why Tail Bounds?

Observation

Exact tail probabilities are often hard to compute.

Exact Computation

- Binomial sum
- Poisson infinite sum

These are complicated and long.

Idea

Instead of computing exact probability:

$$P(X \geq k)$$

We compute an upper bound:

$$P(X \geq k) \leq \text{something simple}$$

Key Idea

Approximate safely instead of computing exactly.

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5. Running Example

Experiment

Flip a fair coin n times.

Distribution

$$X = \text{number of heads}$$

$$X \sim \text{Binomial}(n, 1/2)$$

Mean

$$E[X] = n/2$$

Question

What is P(X 3n/4)?

Interpretation

This represents getting unusually many heads, which lies in the tail region.

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6. Markov's Inequality

Key Idea

- Uses only the mean
- Large values are rare

Statement

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Assumption

$$X \geq 0$$

Proof (Detailed Steps)

1. Start from expectation:

$$E[X] = \sum xP(X = x)$$

2. Consider only terms where x a:
Since x a,

$$xP(X = x) \geq aP(X = x)$$

3. Therefore:

$$E[X] \geq \sum_{x \geq a} aP(X = x)$$

4. Factor out a:

$$E[X] \geq a \sum_{x \geq a} P(X = x)$$

5. Recognize probability:

$$\sum_{x \geq a} P(X = x) = P(X \geq a)$$

6. Hence:

$$E[X] \geq aP(X \geq a)$$

7. Rearranging:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Example Logic

If average = 10, then values like 100 are rare.

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7. Chebyshev's Inequality

Statement

$$P(|X - \mu| \geq k) \leq \frac{\mathrm{Var}(X)}{k^2}$$

Proof (Step-by-Step)

1. Let Y = (X -)²

2. Apply Markov inequality:

$$P(Y \geq k^2) \leq \frac{E[Y]}{k^2}$$

3. Substitute:

$$P((X - \mu)^2 \geq k^2) \leq \frac{E[(X - \mu)^2]}{k^2}$$

4. Recognize variance:

$$E[(X - \mu)^2] = \mathrm{Var}(X)$$

5. Therefore:

$$P(|X - \mu| \geq k) \leq \frac{\mathrm{Var}(X)}{k^2}$$

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8. Chernoff Bound

Key Idea

Use exponential transformation.

Steps

1. Start with probability:
$$P(X \geq a)$$
2. Apply monotonic transformation:
$$P(e^{tX} \geq e^{ta})$$
3. Apply Markov inequality:
$$P(e^{tX} \geq e^{ta}) \leq \frac{E[e^{tX}]}{e^{ta}}$$
4. Therefore:
$$P(X \geq a) \leq \frac{E[e^{tX}]}{e^{ta}}$$

Insight

- Converts problem into expectation of exponential
- Leads to tighter bounds

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9. Poisson Tail Bound

Used for Poisson distributions using Chernoff method.

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10. Jensen's Inequality

Convex Function Definition

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$$

Statement

$$f(E[X]) \leq E[f(X)]$$

Interpretation

Function of expectation is less than expectation of function.

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11. Summary

- Tail = extreme event probability
- Exact computation = hard
- Bounds = easier

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12. Comparison

- Markov → uses mean
- Chebyshev → uses variance
- Chernoff → strongest bound

End of Complete Exam Scribe